Unless explicitly requested by a problem, do not include sketches as part of your proof. You are free to use the result from any problem on this (or previous) assignment as a part of your solution to a different problem even if you have not solved the former problem.

Problem 1 (2pts)**.** Let G be a regular graph. Show that $\chi'(G) = \Delta(G)$ if and only if G is 1-factorable.

Solution. Let $n = |V(G)|$. Thus, if M is a matching, then $|M| \leqslant \frac{n}{2}$, with equality if and only if M is perfect. Since G is regular, all vertices have the same degree, say r. Thus $\Delta(G) = r$. Also, by the handshaking lemma, $|E(G)| = \frac{nr}{2}$.

($\Rightarrow$) Let $t = \chi'(G)$, so by definition we have a partition of the edge-set into t matchings, say

$$E(G) = M_1 \sqcup \cdots \sqcup M_t.$$

If $\chi'(G) = \Delta(G)$, then the partition above becomes $E(G) = M_1 \sqcup \cdots \sqcup M_r$. If some M_j has $|M_j| < \frac{n}{2}$, using that all other M_i's have $|M_i| \leqslant \frac{n}{2}$ we get

$$\frac{nr}{2} = |E(G)| = \sum_{i=1}^{r} |M_i| < \frac{n}{2} r,$$

a contradiction. Thus $|M_j| = \frac{n}{2}$ for all $j \in \{1, \ldots, r\}$, thus all M_j's are perfect.

($\Leftarrow$) If G is 1-factorable, we have a partition

$$E(G) = M_1 \sqcup \cdots \sqcup M_k,$$

where k is a non-negative integer and each M_j is a perfect matching. In particular, $|M_j| = \frac{n}{2}$ for each $j \in \{1, \ldots, k\}$. Now we see that

$$\frac{nr}{2} = |E(G)| = \sum_{i=1}^{k} |M_i| = \sum_{i=1}^{k} \frac{n}{2} = k\frac{n}{2},$$

which gives $r = k$. This means that we have a partition of $E(G)$ into r matchings, thus $\chi'(G) \geqslant r = \Delta(G)$ by definition. By Vizing's Theorem, it must be $\chi'(G) = \Delta(G)$. □

Problem 2 (2pts)**.** Let G be a bipartite graph with parts U, W where $|U| = |W| = k \geqslant 1$. Prove that if $|E(G)| > k(k-1)$, then G has a perfect matching.

Hint: König's Theorem $\alpha'(G) = \beta(G)$.

Solution. [#1] Since G is bipartite with parts of size k, to show that G has a perfect matching it suffices to show that there is a size-k matching, or equivalently, that $\alpha'(G) \geqslant k$. Now, $\alpha'(G) = \beta(G)$ by König's Theorem, so we only need to show that $\beta(G) \geqslant k$.

Let $C \subseteq V(G)$ be a minimum vertex cover of G, so $\beta(G) = |C|$. Since G is bipartite with parts of size k, we have $\deg v \leqslant k$ for all $v \in V(G)$. For a vertex v, let E_v denote the set of edges incident to v, thus $|E_v| = \deg v$. Now, since the vertices in C cover all edges of G, we have

$$k(k-1) < |E(G)| = \left| \bigcup_{v \in C} E_v \right| \leqslant \sum_{v \in C} |E_v| = \sum_{v \in C} \deg v \leqslant \sum_{v \in C} k = |C|k = \beta(G)k.$$

Thus $k - 1 < \beta(G)$, which gives $k \leqslant \beta(G)$, as desired. □

Solution. [#2] Let H denote the complete bipartite graph with both parts of size k. Thus G is a spanning subgraph of H. Moreover, H has $|E(H)| = k^2$. Thus the hypothesis $|E(G)| > k(k-1) = k^2 - k$ says that we can get G from H by deleting fewer than k edges (and no vertices).

Since H is a k-regular bipartite graph, Theorem 8.15 (or HW 10.3) tells us that we can partition

$$E(H) = M_1 \sqcup \cdots \sqcup M_k$$

where each M_i is a perfect matching for H.

Now, since we can get G from H by deleting fewer than k edges, one of the k matchings M_i must remains un-touched, i.e., $M_i \subseteq E(G)$. Thus M_i is also a perfect matching for G. □

Problem 3 (2pts)**.** Let G be a bipartite graph with parts U, W and fix an integer $k \geqslant 1$. Let $U = U_1 \sqcup \cdots \sqcup U_k$ and $W = W_1 \sqcup \cdots \sqcup W_k$ be any partitions (some of the U_i's or W_j's may be empty). Note that $|U|$ and $|W|$ have nothing to do with k; k is just the number of pieces in each partition.

Prove that if $|E(G)| < k$, then there is a bijection $\pi\colon \{1, \ldots, k\} \to \{1, \ldots, k\}$ such that G has no edges between U_i and $W_{\pi(i)}$ for each $i \in \{1, \ldots, k\}$.

Hint: construct a bipartite graph using the U_i's and W_j's as *vertices*, and use Problem 2.

Solution. Build a bipartite graph H with parts $A = \{U_1, \ldots, U_k\}$ and $B = \{W_1, \ldots, W_k\}$ where $U_iW_j \in E(H)$ iff there are no edges of G between elements of U_i and elements of W_j. Then the desired bijection π exists if and only if H has a perfect matching.

Now, the U_i's and W_j's are disjoint, so for each $e \in E(G)$, the edge e connects exactly one pair U_i and W_j . Some of these edges may connect the same U_i and W_j , but, in any case, there are at most $|E(G)| < k$ many **non**-edges in H between A and B. Thus

$$|E(H)| > k^2 - k = k(k-1).$$

Hence H has a perfect matching by Problem 2. □

Problem 4 (2pts)**.** Let $g \geqslant 2$ be an integer and let G be a connected plane graph on n vertices wherein every face is bounded by a cycle of G. Prove that if G has no cycles of length g or smaller, then

$$|E(G)| \leqslant \frac{g+1}{g-1}(n-2).$$

Solution. Since each face of G is bounded by a cycle and G has no cycles of length $\leqslant g$, we must have $\text{len}(f) \geqslant g+1$ for all $f \in F(G)$. Thus, the headshaking lemma yields

$$2|E(G)| = \sum_{f \in F(G)} \text{len}(f) \geqslant \sum_{f \in F(G)} (g+1) = (g+1)|F(G)| \quad \Rightarrow \quad |F(G)| \leqslant \frac{2}{g+1}|E(G)|.$$

Now, since G is connected, the Euler's formula gives

$$2 = n + |F(G)| - |E(G)| \leqslant n + \frac{2}{g+1}|E(G)| - |E(G)| = n - \frac{g-1}{g+1}|E(G)|,$$

which implies

$$|E(G)| \leqslant \frac{g+1}{g-1}(n-2),$$

as desired. □

Problem 5 (2pts)**.** Prove a special case of the 4-color theorem: If G is a planar, triangle-free graph, then $\chi(G) \leqslant 4$.

Solution. Suppose for the sake of contradiction that $\chi(G) \geqslant 5$ and let H be any 5-critical subgraph of G, which exists by Proposition 3 of Lecture 20. Note H is connected and has $\delta(H) \geqslant 4$ by Proposition 4 of Lecture 20. Additionally, since H is a subgraph of G, we know that H is also planar and triangle-free. Thus, if $n = |V(H)|$, we have $2n - 4 \geqslant |E(H)|$. By the handshaking lemma,

$$2n - 4 \geqslant |E(H)| = \frac{1}{2} \sum_{v \in V(H)} \deg_H v \geqslant \frac{1}{2} \sum_{v \in V(H)} 4 = \frac{1}{2} 4n = 2n,$$

a contradiction. □